Report ML Project 1

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I. Introduction

In this project, we compare six different methods to determine the risk of a person in developing coronary heart disease based or not. We have chosen to compare mean square gradient descent, mean square stochastic descent, ridge regression, logistic regression, least square and regularized logistic regression. We used data from Behavioral Risk Factor Surveillance System, a telephone survey about health. The training set has 328135 samples and the test set has 109379 samples. The data has 321 features. of their personal lifestyle factors.

II. Models and Methods

A. Models

1) Mean squared error gradient descent: Iteratively updates the weights to minimize the MSE using the exact gradient computed over the entire training dataset. The arguments are `--max_iters`, the maximum of iterations, and `--gamma`, the learning rate.

2) Mean squared stochastic gradient descent: Approximates the gradient using a single random data point per iteration (using a fixed seed for exact reproducibility). The arguments are `--max_iters`, the maximum of iterations, and `--gamma`, the learning rate.

3) Least square: Computes the exact analytical solution to minimize the Mean Squared Error (MSE) using the normal equations. This baseline method requires no additional hyperparameters.

4) Ridge regression: Extends the least squares approach to prevent overfitting by adding an L2 regularization penalty. The strength of this penalty is directly controlled by the `lambda_` argument.

5) Logistic regression: Optimizes the logistic loss iteratively. The weight updates are controlled by the gamma step size and run for the specified `max_iters`.

6) Regularized logistic regression: Incorporates an L2 penalty into the logistic loss to handle high-dimensional feature spaces and prevent overfitting. This method utilizes the full set of hyperparameters, using `lambda_`, the regularization parameter, to control the regularization strength, alongside gamma and `max_iters` for the iterative optimization.

B. Pre-processing

The raw features cannot be fed directly to a linear model: many answers are missing, some numbers are survey codes, and most features are categorical. All statistics below (medians, means, standard deviations, categories) are computed on the training set only and then applied unchanged to the test set, to avoid any information leakage.

1) **Special codes.** In the BRFSS codebook, the largest values of a question are not real values: 7/77/777 mean "don't know" and 9/99/999 "refused" (replaced by a missing value), while 88/888 mean "none" (replaced by 0). For instance, $\texttt{PHYSHLTH} = 88$ means 0 days in bad health, not 88. A code is only replaced when it is isolated (the value just below it is absent), so that legitimate values such as $\texttt{EMPLOY1} = 7$ ("retired") are kept.
2) **One-hot encoding.** 219 of the 321 features take at most 10 distinct values and are categorical: their numerical value has no meaning (e.g. marital status). Each is replaced by one binary column per category, a missing answer being its own category, since not answering is often informative. The most frequent category is dropped to avoid an exact collinearity with the bias column.
3) **Missing values.** The remaining numerical features with more than 50% of missing values are dropped; the others are imputed with their median, which is robust to outliers.
4) **Standardization.** Every column is centered and scaled to unit variance, and a bias column of ones is added, giving 774 features. For SGD only, the binary columns are left as 0/1: once standardized, a rare category takes values up to $1/\sqrt{\text{frequency}} \approx 500$ on a few samples, which makes a single SGD step diverge.

Labels and decision threshold. The labels are mapped from $\{-1, 1\}$ to $\{0, 1\}$ for the logistic models. Since only 8.8% of the samples are positive, predicting with the default probability threshold of 0.5 detects very few positives. We therefore choose the threshold that maximizes the F1 score on the training set (≈ 0.2 in probability), and predict a positive label when the score exceeds it.

Ablation study. Fig. 1 shows the 5-fold cross-validated F1 of logistic regression when the steps are added one by one. The special codes (+0.012) and the one-hot encoding (+0.015) bring all the gain; adding missing-value indicators or polynomial features (degree 3) does not improve the score, so they are not used.

C. Tuning

Now we have got our data pre-processed and cleaned, our goal is to find the best hyperparameter. We will use Grid search that consist of trying different hyperparameter to get the best F1-score. For each try, we use K-fold cross-validation, with $k = 5$. It is used to evaluate a model's performance by dividing the dataset into K equal parts. The model is trained on K-1 folds and tested on the single remaining fold, repeating this cycle K times so that every piece of data serves as the test set

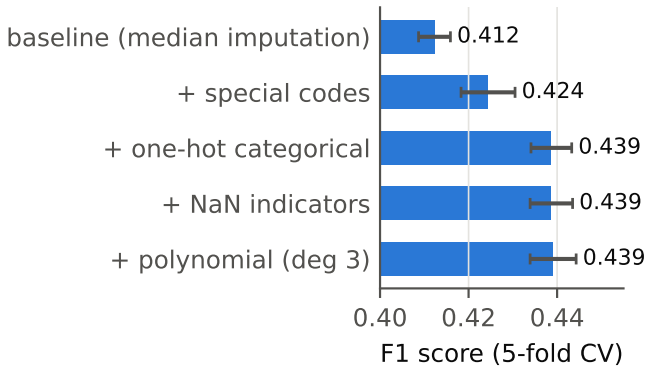


Fig. 1. Ablation study: 5-fold cross-validated F1 score of logistic regression for each pre-processing step (mean ± std over the folds).

exactly once. The final result is given by averaging the results of all iterations. According to the graph 2, we obtain for Ridge regression $\lambda = 10^{-4}$, for regularized logistic regression $\lambda = 10^{-4}$, for MSE on gradient descent $\gamma = 0.05$, on stochastic gradient descent $\gamma = 1e-4$ and for logistic regression $\gamma = 0.5$. We fix maximum of iteration for gradient descent and logistic regression to 1000 and for stochastic gradient descent to 300000. Moreover, we fix for regularized logistic regression γ to 0.5.

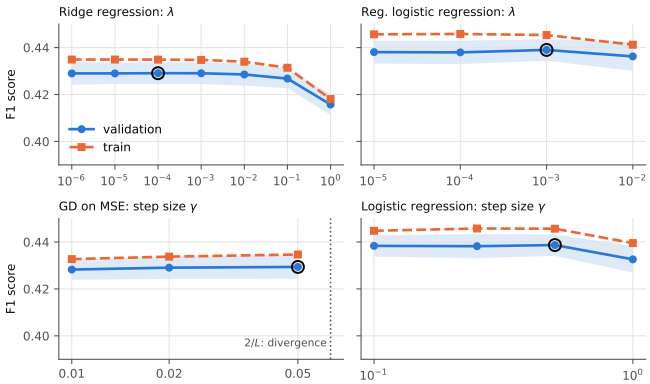


Fig. 2. Validation curves

III. Results

Methods	F1-score	Accuracy	time (sec)
Least squares	0.4291 ± 0.0049	0.8733	87
Ridge regression	0.4291 ± 0.0047	0.8724	76
MSE Gradient descent	0.4294 ± 0.0050	0.8736	669
MSE Stochastic gradient descent	0.4202 ± 0.0055	0.8682	166
Logistic regression	0.4387 ± 0.0046	0.8782	725
Ridge logistic regression	0.4380 ± 0.0050	0.8778	766

TABLE I
RESULTS WITH F1-SCORE AND ACCURACY WITH 5-FOLD CROSS-VALIDATION

Methods	F1-score	Accuracy
Least squares	0.439	0.877
Ridge regression	0.438	0.877
MSE Gradient descent	0.437	0.876
MSE Stochastic gradient descent	0.430	0.869
Logistic regression	0.450	0.883
Regularized logistic regression	0.451	0.881

TABLE II
RESULTS WITH F1-SCORE AND ACCURACY ON THE AICROWD TEST SET

IV. Discussion

A. discussion of results

According to the result, I, the best performance in cross-validation are given by the logistic regression. However, the counterpart is the resources. It takes around 725 seconds to train. A good alternative is ridge regression. Within 76 seconds, we obtain a similar accuracy and F1-score. On the test set, the best accuracy is reached by logistic regression, with 0.883. The best F1-score is given by regularized logistic regression, with 0.451.

The low F1-score is explained by the low percentage of positive labels on the training set, only 8.8%. According to table II, it is very hard for the model to detect minority case. After comparison between train and test set, we can conclude that models don't overfit. Moreover, the results on the test set seem better on the training set. We can explain it because the final model is trained on the whole training set instead of 80% of it in each fold, and the public leaderboard is computed on a subset of the test set only.